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Artificial Intelligence for Speech Classification and Enhancement of Speech and Language Disorders: Techniques, Applications, and Future Directions

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ABSTRACT Artificial intelligence (AI) has advanced the classification and management of speech and language disorders, which are conditions characterized by the progressive deterioration of speech, language, and communication abilities. Speech production relies on coordinated respiratory, phonatory, articulatory, and cognitive mechanisms. Disruptions in these systems, particularly due to neurodegenerative diseases such as Parkinson's disease, Alzheimer's disease, and dysarthria, can lead to speech impairments affecting clarity, fluency, and timing. This systematic literature review synthesizes insights from 231 studies (2007–2024) on machine learning (ML) and deep learning (DL) techniques for classifying and analyzing 16 distinct speech and language disorders, emphasizing neurodegenerative conditions. This review highlights the progression from traditional ML approaches, such as support vector machines, to advanced DL architectures, including convolutional and recurrent neural networks, and explores the potential of hybrid and multimodal models. Feature extraction methods—acoustic, prosodic, and linguistic—are analyzed for their ability to capture speech complexities, whereas multimodal frameworks that integrate acoustic, linguistic, and physiological signals show promise for increased diagnostic accuracy. In addition to classification, emerging work has focused on deep learning-based speech enhancement methods aimed at improving intelligibility and signal quality for pathological speech. Techniques such as multi-scale feature fusion and attentive recalibration network (MFFR-net), noise-aware variational autoencoders, LLM (Large language models)-integrated communication aids, and ultrasound-guided reconstruction for impaired articulation are enhancing the clarity of speech inputs for downstream diagnostic tasks. These approaches are especially beneficial for noisy environments and low-resource users, providing improved preprocessing pipelines for classification systems. The key challenges identified include the lack of longitudinal and cross-cultural datasets and the limited explainability of artificial intelligence (AI) models in clinical contexts. LLMs and adaptive techniques are identified as opportunities to enable disorder-agnostic frameworks and personalized care. Future research should prioritize multimodal feature integration, disorder-specific modeling, joint modeling of enhancement and classification and the creation of more comprehensive datasets to increase the accuracy, generalizability, and clinical adoption of AI-based speech and language disorder classification systems.

INDEX TERMS Classification, Speech Enhancement, Feature extraction, Speech and language disorders, Systematic review

I. INTRODUCTION

Speech and language disorders refer to difficulties in communication, but they differ in nature. Language

disorders involve problems in understanding (receptive) or producing (expressive) language, affecting grammar, meaning, and other linguistic aspects. Speech disorders, on the other hand, involve difficulties in producing speech

sounds correctly, including issues with voice, articulation, and fluency. While language disorders affect both spoken and written communication, speech disorders mainly impact the vocal expression of language [1], [2]. Frequently observed disorders include stuttering, apraxia of speech, dysarthria, phonological disorders, aphasia, and various voice disorders. The etiology of these disorders may involve genetic predispositions, neurological dysfunctions, physical trauma, or developmental anomalies [3], [4], [5]. Neurodegenerative conditions such as specific language impairment (SLI), Parkinson's disease, Alzheimer's disease, and amyotrophic lateral sclerosis markedly hinder communicative ability and diminish quality of life by disrupting both motor and cognitive systems, thereby resulting in articulation impairments, dysarthria and speech-related challenges [6]. Individuals with these conditions may encounter substantial challenges in articulating words, sustaining fluent speech, or modulating vocal qualities, which adversely affects their social interactions, educational development, and overall quality of life.

In parallel with the growing clinical challenges, there has been a marked global increase in the demand for assistive technology (AT). Assistive technology (AT) is an umbrella term encompassing assistive products, their related systems, and services. Assistive products help maintain or improve an individual's functioning related to cognition, communication, hearing, mobility, self-care, and vision, thus enabling their health, well-being, inclusion, and participation [7]. The World Health Organization estimates that approximately 2.5 billion individuals globally require assistive technology (AT). Projections suggest that this figure may increase to approximately 3.5 billion by 2050 [7]. However, access to appropriate and specialized speech-related AT remains limited in many settings. Proficient assessment and classification of speech and language disorders are pivotal for prompt diagnosis and intervention, thereby assisting individuals in the restoration of communicative capabilities and enhancement of their daily functional activities [8]. A study conducted in China analyzed the service delivery and satisfaction levels of assistive devices among 872 individuals with disabilities via the World Health Organization's Rapid Assistive Technology Assessment (rATA) questionnaire and revealed that AI-enhanced assistive solutions have demonstrated promising outcomes in improving the accuracy, accessibility, and personalization of diagnostic and intervention tools, enhancing the communicative abilities and autonomy of affected individuals [9].

Artificial intelligence (AI) has introduced transformative possibilities for addressing speech and language disorders and developing assistive tools. Traditional methods of speech assessment have typically relied on clinical evaluations conducted by speech-language pathologists [9]. However, these processes can be subjective, time-consuming, and imprecise. To address these challenges, automated speech assessment tools based on machine learning (ML) [10], [11] and deep learning (DL) models [12]

have evolved, providing more objective, scalable, and efficient solutions. These tools often rely on biomedical signal processing techniques to extract meaningful acoustic and physiological information from speech data. Machine learning could enable data to be used more efficiently to provide insights into disease mechanisms and to help with earlier diagnosis, prognosis, patient stratification and development of new therapies [13].

In addition to classification, speech enhancement techniques have emerged as a complementary avenue to address signal distortion and improve intelligibility before diagnosis. Recent research has introduced noise-aware deep learning models [14], multi-scale fusion architecture (e.g., MFFR-net) [15], and neuromorphic or low-power transformer-based systems [16] that aim to refine degraded or pathological speech. For instance, models like Noise-Aware VAE, LangChain-enhanced AAC devices [17], and ultrasound-tongue-based reconstruction networks [18] enhance disordered speech signals, enabling more reliable input for subsequent classification or therapeutic interpretation. These enhancement techniques contribute not only to assistive communication but also to more robust downstream applications in AI pipelines, particularly in real-world noisy conditions or with low-quality input from impaired speakers.

However, the efficacy of automated tools [8], [19] is determined by the model's ability to handle a range of disorders and adapt to different speech features [20], [21] and linguistic contexts. These tools employ sophisticated algorithms to analyze vocal characteristics, including acoustic, prosodic, and linguistic components, thus enabling the precise categorization of different speech and language disorders. The classification of speech and language disorders is critical not only for identifying a particular disorder but also for developing customized therapeutic strategies. Recent developments in multimodal AI frameworks that integrate features from different modalities, including acoustic signals, linguistic cues, and physiological data, offer a unique approach path for addressing the complexities related to neurodegenerative speech and language disorders. Loss of speech after paralysis is devastating, but circumventing motor pathway injury by directly decoding speech from intact cortical activity has the potential to restore natural communication and self-expression [22].

Despite the considerable global prevalence of conditions such as multiple sclerosis (MS) (~35–100 per 100,000) [23], Lewy body dementia (LBD) (~1% of individuals over 65) [24], and myasthenia gravis (~14–20 per 100,000) [25], research focusing on speech manifestations is disproportionately scarce in comparison with that of individuals with disorders such as Parkinson's disease. Moreover, gaps remain in dataset diversity, generalizability, and real-world applicability of AI-driven assistive solutions.

To bridge these gaps, this study offers a comprehensive overview of research on speech and language disorders, employing diverse AI techniques for diagnosis and monitoring through voice samples. Additionally, this review incorporates recent advances in speech enhancement, examining how signal processing and deep neural methods are being used to improve disordered speech quality prior to classification. By offering a comprehensive synthesis of the current state of research on various speech and language disorders and classification methods, this review contributes to a better understanding of the advancements and challenges associated with the automated classification and enhancement of speech and language disorders. We aim to address the following research questions:

RQ1: What speech and language disorders have been addressed by AI models, including neurodegenerative conditions?

RQ2: What corpora are used, and how do they support diverse, longitudinal, and multimodal analyses?

RQ3: What methods, feature selection methods, and metrics are used for classifying different speech and language disorders?

RQ4: What are the real-world applications, limitations and future opportunities for improving AI-based speech and language disorder classification?

RQ5: What are the recent advancements in speech enhancement methods supporting downstream classification tasks?

This paper makes several key contributions to the literature on artificial intelligence (AI) applications for speech and language disorder diagnosis, classification and enhancement. First, it synthesizes findings from a broad corpus of studies, providing a comparative analysis of AI applications across neurodegenerative and other speech-affecting conditions. This includes both classification-focused models and enhancement techniques that improve speech intelligibility prior to analysis. Second, it identifies critical research gaps, emphasizing the limited focus on certain high-prevalence conditions despite their clinical importance as well as the underutilization of enhancement frameworks in clinical pipelines. Third, the study highlights challenges in dataset availability, diversity, and longitudinal scope, suggesting the need for more comprehensive resources to address these limitations. Finally, it explores advancements in AI methodologies, emphasizing the need for personalized, explainable, and clinically integrated models to address the unique complexities of speech and language disorders.

The remaining sections of this paper are organized as follows. The Background section highlights the need for this literature review by comparing it with the previous studies. The speech and language disorder classification architecture outlines a foundational framework for classifying such disorders. The methodology section details the systematic literature review (SLR) process, including inclusion criteria and thematic analysis. The Results section examines different speech and language disorders based on their

prevalence within the realm of AI research, analyzes the datasets and methodologies applied, and explains emerging themes and shortcomings. The Discussion section correlates the findings with both theoretical and practical implications, suggesting strategies that could mitigate the identified gaps. The Conclusion summarizes the contributions of the study and delineates potential research directions, emphasizing the importance of better AI frameworks for clinical applications in the management of speech and language disorders.

II. BACKGROUND

In recent years, the application of artificial intelligence (AI) and machine learning (ML) techniques for identifying speech and language disorders has increased, particularly through advances in automatic speech recognition (ASR) systems. ASR has evolved with the adoption of deep learning architectures, enabling improved robustness in noisy, real-world, and low-resource environments. Kheddar et al.[26] provided a survey on the role of deep transfer learning in ASR, demonstrating how pre-trained models can be fine-tuned to improve performance in low-resource, mismatched, and speaker-variable conditions, thus enhancing generalization across domains. Complementing this, Kheddar et al.[27] provided a comprehensive survey of advanced deep learning approaches in ASR, emphasizing the use of transformers, deep reinforcement learning, and federated learning to enhance speech recognition accuracy under varying conditions. The study[28] provides a comparative analysis of three prominent speech recognition frameworks—SpeechBrain, OpenAI's Whisper, and Facebook's Wav2Vec2—evaluating their performance across various datasets and languages. This work is particularly relevant for speech disorder research as it highlights the models' generalization capabilities, robustness to noise, and potential suitability for clinical and assistive applications involving impaired speech. This section highlights the importance of this literature review relative to prior studies, emphasizing its broader scope and the unique contributions it makes beyond existing reviews.

Heatmap in Fig.1 compares speech and language disorders addressed by this work with those reported in prior review articles. Brahmi et al. [29] reviewed different ML assistive technologies for speech and language disorders. Hamza et.al.[30] focused on machine learning frameworks for dysarthria severity classification, analyzing feature extraction, model performance, and benchmark comparisons on clinical datasets such as TORGO and UA-Speech. Thirty-three studies on speech-based deep learning approaches for Parkinson's disease classification published between January 2020 and March 2024 reviewed and discussed the limitations of publicly accessible speech-based PD datasets due to privacy concerns [31]. A review by Lounis et al. [32] highlights the growing interest in using advanced techniques such as deep neural networks (DNNs) for mispronunciation detection and diagnosis (MDD) in language learning and highlights the need for diverse and comprehensive datasets,

especially for tonal languages such as Mandarin. Across these fields, one of the most pervasive challenges is the lack

of comprehensive and standardized datasets, particularly for uncommon or specialized voice

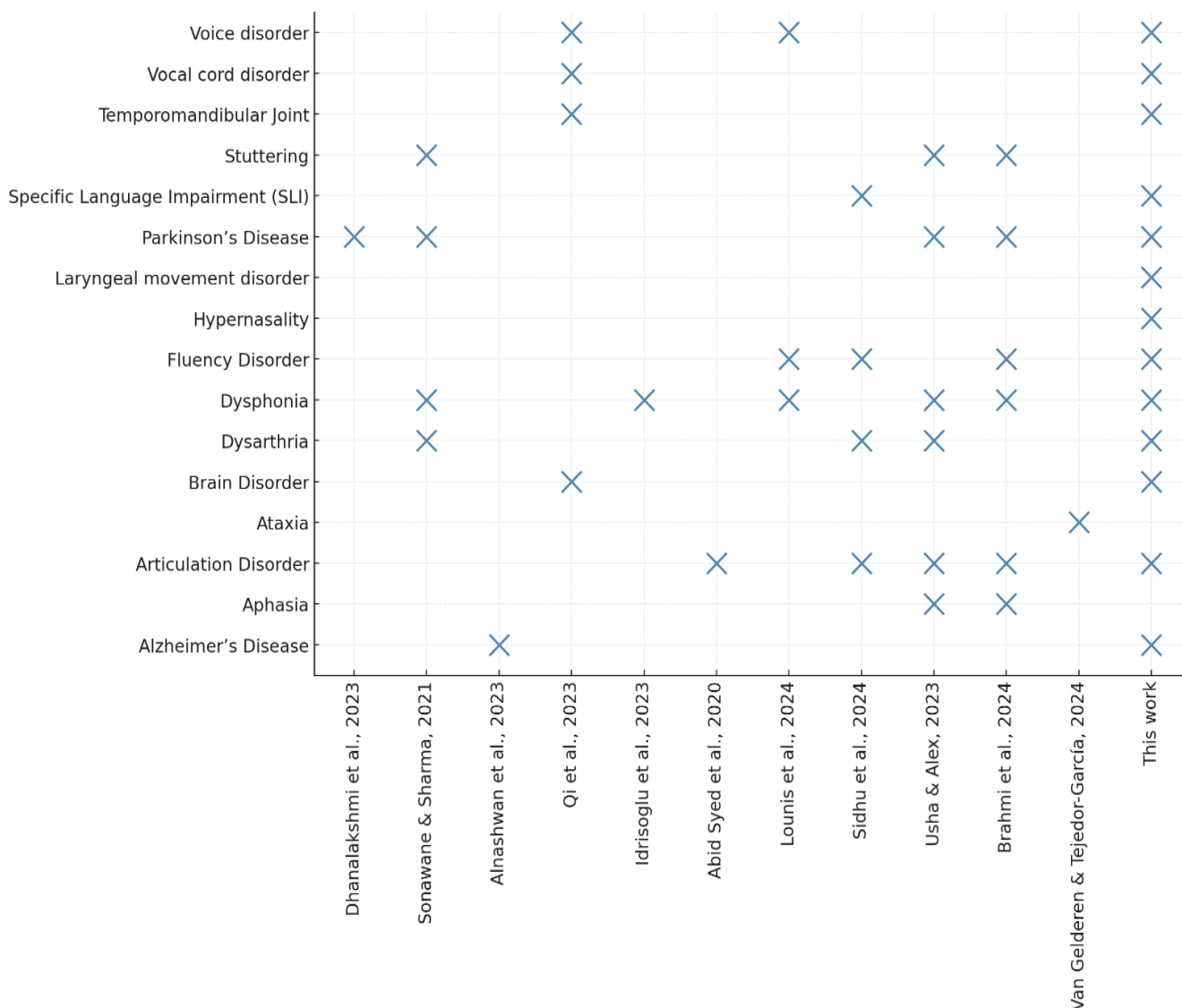


FIGURE 1. Speech and language disorders from review articles

disorders. The study performed by Sindhu et al. [20] emphasized the absence of universally accepted assessment tools, and the scarcity of studies applying unsupervised methods hinders progress in speech and disorder detection research. Moreover, studies on speech assessment in children, particularly infants under 24 months, highlight the difficulty of early diagnosis [8], with future work necessitating the development of automated tools for quantifying speech patterns to improve diagnostic efficiency. Parkinson's disease (PD) has been a significant focus in [33], with research highlighting the development of intelligent systems to detect and monitor the disease through different algorithms. In a systematic review, Sonawane et al. [34] reinforced the role of these systems, especially in detecting voice disorders related to PD, such as dysphonia and dysarthria, pointing toward the need for automated tools

owing to the limitations associated with manual and subjective evaluation. However, the current literature on predicting PD severity through speech analysis remains limited and requires more advanced machine learning techniques to fill this gap [34].

Recent advancements in stuttering detection with AI were reviewed by Alnashwan et al. [35], but many challenges still need to be solved due to the unavailability of diverse datasets and class imbalance, which significantly affects model generalizability. According to researchers, in existing methods, essential speech features are often altered, and more sophisticated augmentation techniques must be used to increase accuracy in real-world applications [35]. Another growing area of research is the detection of Alzheimer's disease (AD) from spontaneous speech. The gap identified across multiple studies reviewed by Qi et al. [36] is that

longitudinal datasets are needed to track the progression of the disease. Current systems have focused mainly on short recordings, which limit their effectiveness in real-time AD detection. Moreover, [36] emphasized the integration of multimodal data with interpretable neural networks to improve the dependability of the model. Moreover, the review by Idrisoglu et al. [37] focused mainly on individual disorders, with limited research on unified frameworks that handle overlapping symptoms across multiple disorders. The relevance of using voice data for broader disorder detection has also been underlined, emphasizing the inclusion of voice-affecting disorders such as dementia, multiple sclerosis, and amyotrophic lateral sclerosis [37]. Within the field of voice disorders unrelated to neurological diseases, the primary limitation is the overreliance on supervised learning techniques, with the underutilization of available databases such as the Arabic Voice Pathology (AVPD) database. After a meta-analysis of 45 studies on voice disorders, Syed et al. [38] reported that despite many datasets suffering from imbalance and a lack of global representation due to language restrictions, machine learning algorithms, especially support vector machines (SVMs), were common. The importance of CNN and RNN architecture is discussed in [39].

In addition to classification, this review also encompasses recent advancements in speech enhancement—a crucial aspect of assistive communication tools. Deep learning-based speech enhancement models have shown significant potential in recovering intelligible speech from noisy or distorted signals, making them valuable in both clinical and real-world environments. As reported by [40], the evolution of deep learning architectures—from traditional Deep Neural Networks (DNN) to convolutional neural networks (CNN) and Generative Adversarial Networks (GANs)—has outperformed classical methods like Wiener filtering or spectral subtraction in handling non-stationary noise. Similarly, [41] emphasizes that modern speech enhancement methods must balance intelligibility and naturalness, with neural models offering promising directions but still lacking phonetic specificity, such as formant preservation. The review by [42] underscores the value of multimodal and attention-based architectures, particularly those leveraging transformer models, in refining speech signals across diverse acoustic conditions. Nevertheless, these studies tend to remain theoretical or noise-focused, often overlooking the specific challenges posed by disordered speech.

More recent contributions have attempted to address these limitations by contextualizing speech enhancement within clinical and assistive domains. The review article [43] explores the use of Convolutional Recurrent Neural Network (CRNNs) and attention-based recalibration techniques for low-latency, low-power applications, including preliminary work on speech-impaired populations. While promising, the lack of clinical benchmarking limits its practical impact. A comparative evaluation of various DL architectures, with specific attention to speaker-independence and medical applicability was identified in [44]. Despite its strengths, the

study remains reliant on synthetic datasets. Collectively, these reviews underscore the rapid progress in DL-based speech enhancement while also illuminating the need for clinically validated models, ethically sound deployment, and inclusive dataset development—particularly when targeting individuals with speech and language disorders. Therefore, by including both classification and enhancement approaches in our analysis, this survey offers a more holistic perspective on the use of AI in the domain of speech and language disorders, supporting both accurate diagnosis and improved communication efficacy.

Table 1 in supplementary material A compares the existing review articles with this survey based on their methodologies. While previous studies provide valuable insights, this survey distinguishes itself from previous reviews by offering a comprehensive analysis of 16 distinct speech and language disorders. Unlike earlier studies that focused primarily on individual disorders such as Parkinson's disease or Alzheimer's disease, this review provides a broader perspective on AI-driven speech disorder classification. This review traces the evolution of AI methodologies, highlighting the transition from traditional machine learning techniques such as support vector machines (SVMs) and random forests (RFs) to more advanced deep learning architectures, including CNNs, recurrent neural networks (RNNs), and transformer-based models. Additionally, the survey emphasizes the significance of multimodal frameworks by integrating acoustic, linguistic, and physiological signals, setting it apart from prior studies that often rely on a single feature set for classification. Table 2 in supplementary material B gives a summary of widely used datasets and associated evaluation metrics in the previous studies. This review critically examines data set limitations, identifying gaps in longitudinal and cross-cultural data availability and advocating for the development of standardized datasets that enhance model generalizability.

Unlike previous reviews that have summarized existing methodologies, the survey identifies critical research gaps and proposes concrete future directions, such as enhancing multimodal feature integration, developing inclusive and representative datasets, and leveraging large language models (LLMs) to improve disorder classification. By providing a forward-looking perspective, this review not only consolidates current research but also lays the groundwork for the development of more effective, personalized, and ethically responsible AI-driven solutions for speech disorder classification.

Previous studies have highlighted the role of artificial intelligence as an emerging form of assistive technology for individuals with speech and language disorders. The potential of AI to enable personalized assessment and treatment for people who stutter has been identified [32]. In the pediatric context, automated speech tools can outperform traditional evaluations, supporting early assistive interventions [8]. Speech-based methods are cost-effective, noninvasive assistive technologies for managing Parkinson's

disease and Alzheimer's disease, respectively [29], [39]. In parallel, recent literature on speech enhancement has advanced the development of intelligent pre-processing modules that improve speech intelligibility and quality in noisy, resource-constrained, or pathological speech settings. Reviews by [42], [40], and others have underscored the importance of deep learning architectures—such as CRNNs, GANs, and Stacked Encoder-Decoder Architecture (SEGAN)—in enabling real-time and low-power enhancement pipelines that can be embedded in assistive

systems. These enhancement models not only facilitate clearer speech perception for listeners and AI systems but also improve the accuracy of downstream classification tasks. While prior reviews largely focus on classification or detection, this work uniquely integrates the growing body of enhancement literature, emphasizing its role in building robust, end-to-end AI-driven assistive technologies capable of supporting a broader range of clinical and functional impairments.

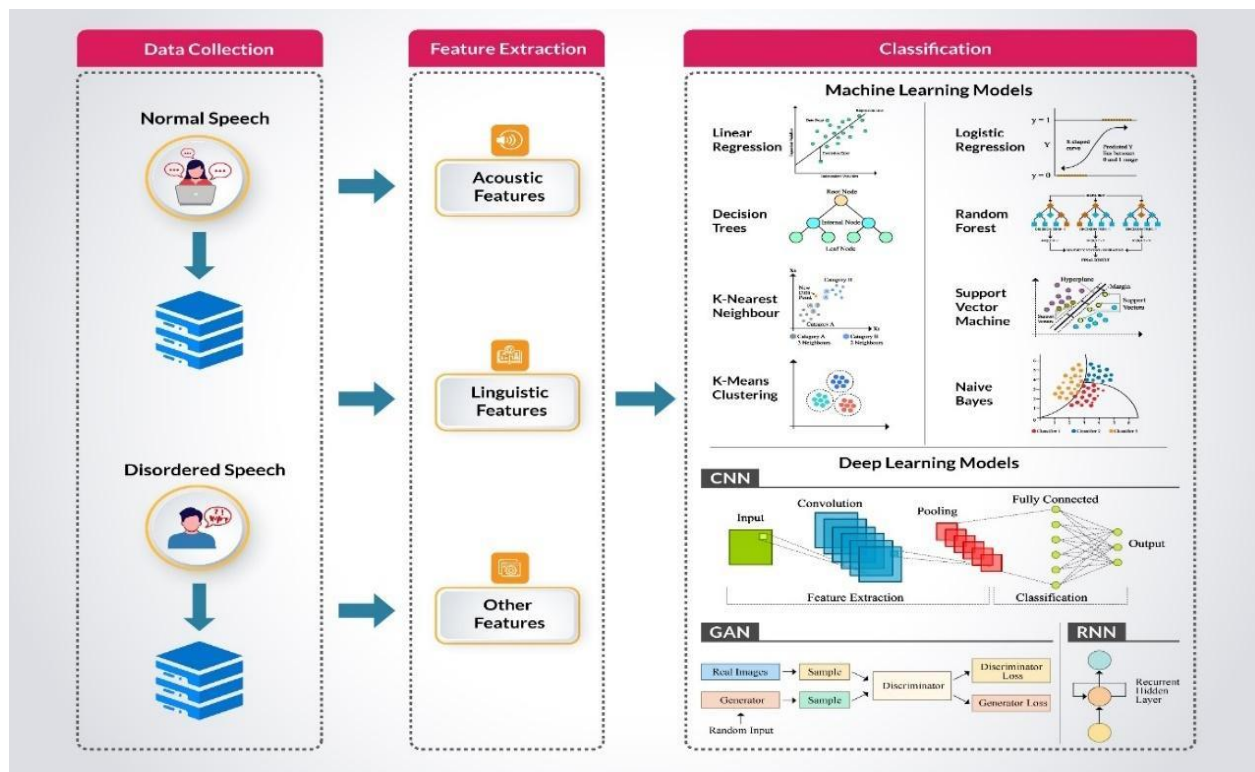


FIGURE 2. Architecture of speech and language disorder classification

III. SPEECH AND LANGUAGE DISORDER CLASSIFICATION ARCHITECTURE

A typical comprehensive speech and language disorder classification model [36] has a three-stage architecture: data collection, feature extraction, and classification, as shown in Fig. 2. In the data collection stage, diverse speech samples from individuals with and without speech and language disorders are gathered. The datasets gathered must cover various disorders, such as stuttering, dysarthria, and apraxia, to capture the unique characteristics of each condition. Speech recordings are commonly complemented with multimodal data [45], such as video for facial movements or physiological data, to capture the full range of disordered speech behaviors. Multimodal data further allow deep insight into complex patterns of speech, thus justifying robust models for real applications. After the data are

collected, preprocessing techniques such as noise reduction, volume normalization, and segmentation are applied to enhance the input for later stages [46]. Speech recordings, along with multimodal data such as video for facial movements or physiological signals, can improve model robustness for real-world applications [45].

The feature extraction stage transforms the raw speech data into a numerical format that can be interpreted via machine learning algorithms. Commonly used features include Mel-frequency cepstral coefficients (MFCCs) [20], pitch, formants, and prosodic elements such as rhythm and intonation [20]. These features capture the acoustic [47] and temporal [48] properties of speech that are disrupted in various disorders. Once the features are extracted, they are fed into the classification stage, where machine learning (ML) or artificial intelligence (AI) models such as support

vector machines (SVMs) [49], random forests [50], or more advanced deep neural networks (DNNs) [51] are employed. Deep learning techniques, particularly convolutional neural networks (CNNs) for spectrogram analysis [52] and recurrent neural networks (RNNs) for temporal speech patterns [53], have proven effective in classifying different types of speech and language disorders. The classification stage assigns labels to the speech data, identifying both whether a speech and language disorder is present and, if so, specifying the type of disorder.

IV. METHODOLOGY

A systematic literature review (SLR) is an independent academic method that aims to identify and evaluate all relevant literature on a topic to derive conclusions about the question under consideration [54]. "Systematic reviews are undertaken to clarify the state of existing research and the implications that should be drawn from this" [55]. The summarization process encompasses the identification, compilation, evaluation, and synthesis of high-quality research evidence within the purview of the research inquiry by adhering to a predetermined protocol [56]. The objective of an SLR is to provide insight into contemporary research so that policymakers may capitalize on current knowledge and rectify the deficiencies that can serve as a foundation for novel research. The predetermined protocol delineates the methodology to be adhered to, articulates the research inquiry, and encompasses information regarding inclusion/exclusion criteria and quality evaluation [54]. All the information pertaining to the methodological approach that steered the implementation of this SLR is predicated on the prespecified SLR protocol [37].

A. SEARCH STRATEGY

A search string was formulated through the application of the most frequently utilized terminologies to identify pertinent scholarly articles; the retrieved articles, along with their

references within the domain, were subsequently employed to unearth additional suitable keywords [57], [58]. By incorporating the newly identified keywords into the search string, a thorough search string was established, as shown in Table 3 in supplementary material C. A tailored variant of the search string was employed in databases such as Web of Science (WoS), PubMed, and Scopus to locate all pertinent peer-reviewed primary journal articles published between 2007 and 2024. Exclusion criteria prioritized clinical specificity, eliminating studies without detailed pathological focus to ensure relevance to speech and language disorder classification.

The search string for AI-driven speech and language disorder classification encompasses diverse studies across various domains and methodologies. It comprises terminology related to speech and voice features, speech and language disorders, AI and computational methods, as well as speech processing and applications. RQ1 addresses a spectrum of speech and language disorders, integrating AI-specific terminology. RQ2 includes multimodal features and implicitly refers to longitudinal corpora through phrases such as "automatic assessment." RQ3 reflects a range of computational methods and feature selection tactics using AI-related terminology. RQ4 emphasizes terms that showcase practical applications and prospective enhancements. RQ5 expands the scope by incorporating terminology related to speech enhancement, signal restoration, intelligibility improvement, and denoising, thereby ensuring that enhancement-focused studies are captured alongside classification-oriented research. The search string is comprehensive and inclusive, harmonizing precision with breadth and targeting both foundational and advanced research in AI-driven speech and language disorder classification. The integration of both general and specialized terms guarantee thorough coverage of AI-based speech and language disorder classification and enhancement.

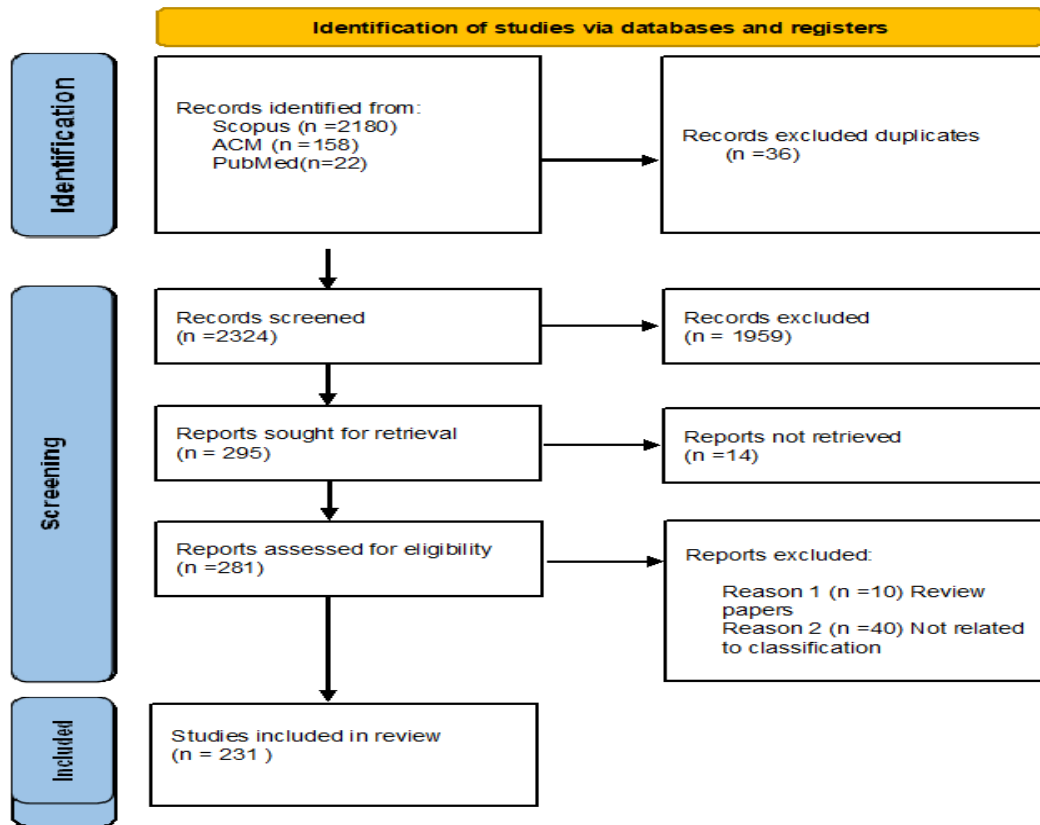


FIGURE 3. PRISMA flowchart

B. STUDY SELECTION PROCESS

The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flowchart [59] pertinent to the present investigation is illustrated in Fig. 3. The automated search yielded a cumulative total of 2360 scholarly articles from the databases, specifically Scopus (2180), Web of Science (158), and PubMed (22). After eliminating duplicate entries, 2324 articles were evaluated through title and abstract screening. Ultimately, 295 manuscripts were incorporated into the full-text reading cohort. A total of fourteen articles could not be accessed. During the comprehensive reading of 281 papers, 50 articles were deemed outside the scope of this research for various reasons: relevance to classification as review papers (n=10), and nonalignment with the classification category (n=40).

Consequently, the final compilation consisted of 231 articles that were utilized for data classification.

Inclusion criteria

- Peer-reviewed Journal articles
- Studies written in English
- Studies published from 2007 onward
- Research aims to diagnose speech and language disorder and speech enhancement.

Exclusion criteria

- A non-peer-reviewed Journal
- Studies written in languages other than English
- Studies published before 2007

V. RESULTS

A. RQ1: SPEECH AND LANGUAGE DISORDERS ADDRESSED

TABLE 1
DISTRIBUTION OF SPEECH AND LANGUAGE DISORDERS IN THE ARTICLES

Disorder Addressed			Conditions	#papers	%
Speech Disorders	Voice Disorders	Dysphonia	Difficulties in pitch, volume, or vocal quality due to vocal fold dysfunction.	9	4.2
		Voice Disorder	Any condition affects vocal sound production, including hoarseness, breathiness, or strain.	50	23.6
		Vocal cord Disorder	Problems in the vocal cords (nodules, polyps, or paralysis) that impair normal voice function.	7	3.3
		Laryngeal movement disorders	Neurological conditions affect the muscles controlling the larynx, leading to voice and breathing issues.	3	1.4
	Fluency	Fluency Disorder	The flow of speech is interrupted, commonly seen in stuttering or cluttering.	2	0.9

	Disorders	Stuttering	Disorder marked by repetitions, prolongations, or blocks that disrupt normal speech flow.	8	3.8
	Speech Disorders	Dysarthria	Disorder due to impaired muscle control due to brain/nerve damage, result slurred/slow speech.	33	15.6
		Articulation Disorder	Individuals have difficulty physically producing certain speech sounds correctly.	8	3.8
		Hypernasality	Resonance disorder where excessive air escapes through the nose during speech, often due to velopharyngeal insufficiency.	1	0.5
		Speech Disorder	Any condition that affects a person's ability to produce sounds that create intelligible speech.	5	2.4
Language Disorders	Language Disorders	Aphasia	Disorder from brain damage (stroke), affecting speaking, understanding, reading, or writing.	5	2.4
		Specific Language Impairment (SLI)	Developmental disorder in which children have difficulty acquiring language despite normal intelligence and no sensory deficits.	6	2.7
Speech/Language Disorders	Neurological or Medical Conditions	Parkinson's Disease	Disorder affects movement and frequently causes speech issues such as dysarthria and hypophonia	63	29.6
		Alzheimer's Disease	Leads to memory loss and cognitive decline, affecting language abilities in advanced stages.	5	2.4
		Brain Disorder	Condition impairs cognitive, speech, or motor functions depending on the region affected.	5	2.4
		Ataxia	Condition impairs coordination and balance, may cause slurred speech due to poor muscle control.	1	0.5
		Temporomandibular Joint	Joint disorder affects jaw movement, lead to speech distortion due to pain or limited mobility.	1	0.5

Understanding the range of speech and language disorders addressed by AI classification methods is crucial for identifying the scope and impact of technological

advancements in speech and language disorder diagnosis and treatment. RQ1 seeks to categorize the disorders and explore how AI technologies have been utilized to identify, analyze, and classify them. A total of 16 disorders, shown in Table 1,

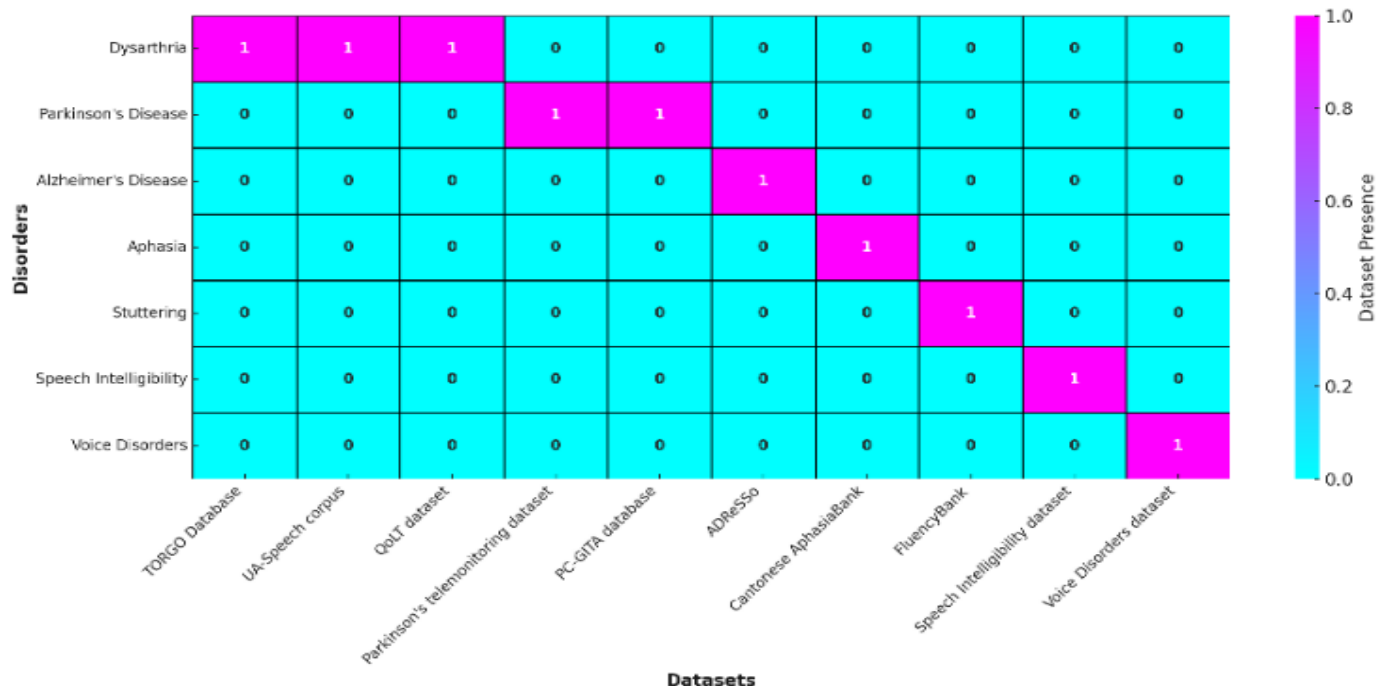


FIGURE 4. Heatmap of speech conditions and datasets

were addressed in this paper, which were classified with machine learning or deep learning models. Most of these

studies focused on Parkinson's disease, dysarthria, and different voice disorders. Distribution of speech and language disorders classified with machine learning or deep

learning models, with Parkinson's disease, dysarthria, and different voice disorders accounting for 68.9% of the total. Interestingly, several high-prevalence conditions, including multiple sclerosis (MS) and Lewy body dementia (LBD), are notably absent in AI research despite their significant impact on speech. MS, with a prevalence of ~35–100 per 100,000 people [23], often causes mixed dysarthria but is overshadowed by more common disorders, such as Parkinson's disease. Similarly, LBD, which affects ~1% of individuals over 65 years of age [60], presents dysarthria and word-finding difficulties but remains underexplored in AI-based speech studies. This disparity underscores the need for more balanced research efforts to address less represented yet clinically significant conditions.

B. RQ2: CORPORA/DATASET

Datasets play a crucial role in the classification of speech and language disorders by providing the foundation for training models, ensuring diversity and variability, enabling accurate labeling, and facilitating research and clinical applications. Furthermore, continuous cooperation between investigators and healthcare professionals is crucial to converting these discoveries into practice, ultimately enhancing patient results and progressing through the domain of speech-language. RQ2 focuses on identifying the types of corpora utilized in AI research, emphasizing their design, diversity, and suitability for longitudinal and multimodal analyses. Table 4 in supplementary material D presents the compilation of the different datasets with their language, modality, and participant description.

Fig. 4 illustrates the relationship between speech and language

disorders and the datasets showing the most applied datasets for specific disorders, such as dysarthria, Parkinson's disease, and stuttering.

Most datasets demonstrate limited linguistic diversity, posing challenges for models aimed at generalizing across languages and dialects. The TORGO database [61] and UA speech database [62] are commonly used for research on dysarthria. These databases include speech recordings from individuals with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS). The Parkinson's UC Irvine Repository and PC-GITA [63] datasets are used to study speech in individuals with Parkinson's disease, often those with dysarthria-related symptoms. In fluency disorders, the SVD

(Saarbruecken Voice Database) [64] has diverse speech samples from individuals with fluency impairments. The LAANA database [65] supports research into language development challenges in children with specific language impairments (SLIs). The Fluency Bank [66], a subset of the TalkBank database, includes annotated recordings for studying fluency patterns, whereas SEP-28k [67] and the German stuttered speech dataset KSoF [68] offer speech samples to train models to recognize and classify stuttering. The MEEI (Massachusetts Eye and Ear Infirmary) Voice Database [69] contains recordings of individuals with various vocal pathologies, including conditions such as hoarseness and vocal nodules.

C. RQ3: AI TECHNIQUES AND METHODS

AI-driven speech and language disorder classification requires advanced computational methods, feature selection techniques, and evaluation metrics. RQ3 identifies algorithms and techniques used for the detection and classification of speech and language disorders based on acoustic, linguistic, and multimodal features. It also analyzes the criteria used for feature selection as well as model performance.

1) FEATURE EXTRACTION AND SELECTION METHODS

Feature extraction converts raw speech into key features for detecting speech and language disorders, whereas feature selection isolates the most relevant features for classification. Features are typically categorized into three primary types: acoustic features derived from audio signals, linguistic features from transcripts, and auxiliary features that further enrich the analysis. Fig. 5 highlights the three primary feature categories and associated methods for effective classification and analysis. SelectKBest [70] is one of the best feature selection methods in machine learning that can be used with acoustic, linguistic, and auxiliary features. It selects the top k features on the basis of statistical tests that assess the relationship between each feature and the target variable. [10] is one of the machine learning models used to rank and select the most important features.

Acoustic features are vital for audio-heavy tasks, whereas linguistic features excel in language structure analysis. Auxiliary methods enhance the overall model precision and computational efficiency.

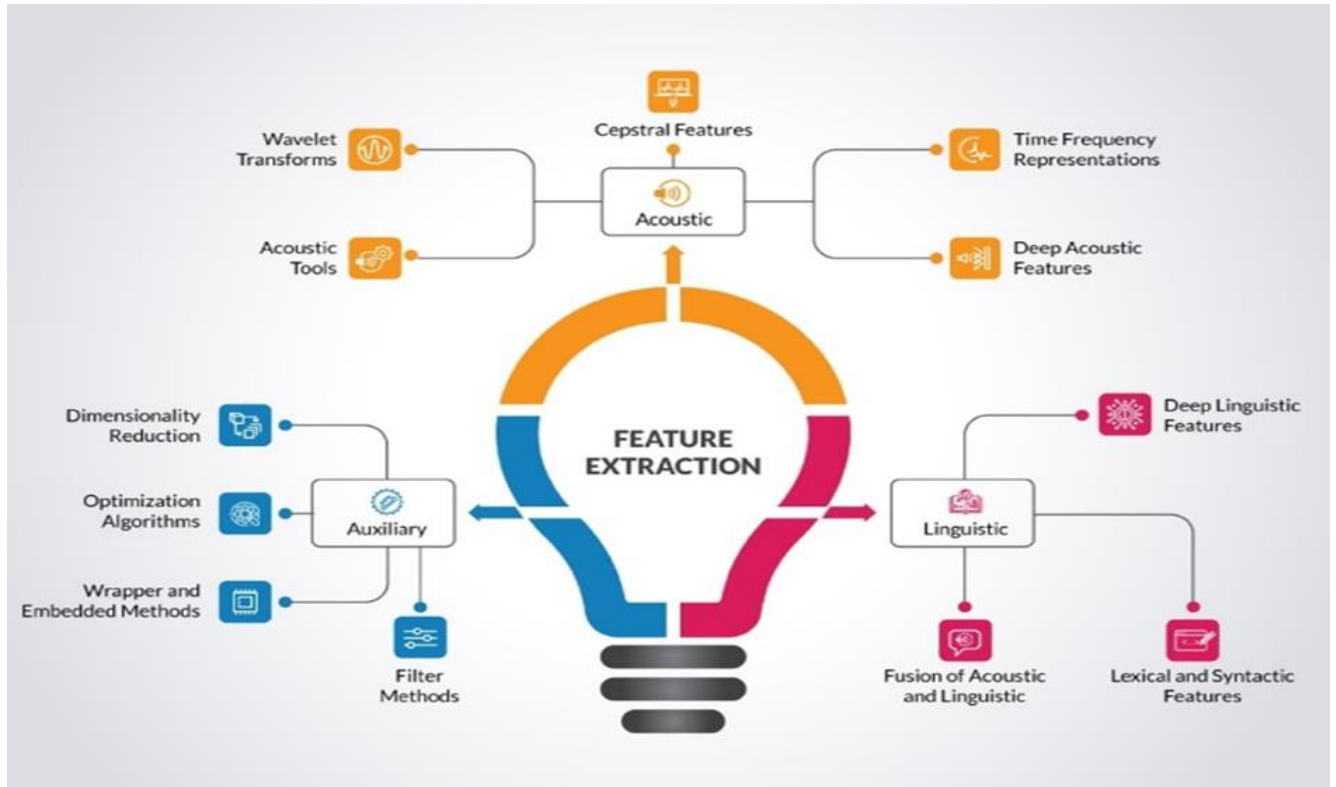


FIGURE 5. Feature extraction and selection in speech and language disorder detection

i) Acoustic features: Acoustic attributes reflect physiological and cognitive changes caused by speech and language disorders, affecting articulation, pitch, fluency, and intelligibility.

- Cepstral features: Mel-frequency cepstral coefficients (MFCCs) [71] and their variants, such as gamma cepstral coefficients (GTCCs) [72] and power-normalized cepstral coefficients (PNCCs) [69], increase sensitivity and noise robustness.
- Wavelet transforms: Methods such as the discrete wavelet transform (DWT)[73] and wavelet scattering transform (WST)[72] analyze signals across multiple resolutions.
- Time–frequency representations: Tools such as short-time Fourier transform (STFT)[61] and mel-spectrograms[74], derived from mel-filter banks, focus on speech-relevant frequencies.
- Acoustic tools such as OpenSMILE (Speech and Music Interpretation by Large-Space Extraction)[49], Praat[75], and Surfboard extract low-level descriptors (LLDs)[76], such as pitch, jitter, shimmer, and the Harmonics-to-Noise Ratio (HNR), for clinical analysis.
- Deep acoustic features: Advanced neural models such as Wav2Vec 2.0 [21], [77] and VGGish [78] extract high-level embeddings that integrate phonetic and prosodic data for sophisticated analysis.

ii) Linguistic features: Linguistic features analyze lexical, syntactic, and semantic aspects, revealing impairments in vocabulary, syntax, and coherence.

- Lexical/syntactic features: Unigrams, bigrams, and trigrams capture structural elements, whereas natural language processing (NLP) models evaluate grammar and sentence patterns.
- Deep linguistic features: Pretrained transformers such as BERT (bidirectional encoder representations from transformers)[49] and BETO (a Spanish variant of BERT) offer embeddings for syntax, semantics, and discourse analysis.
- Feature fusion: Combining acoustic and linguistic data (e.g., MFCCs with BERT embeddings) enhances classification accuracy by creating comprehensive representations.

iii) Auxiliary features: Auxiliary features streamline data, improving accuracy and efficiency through feature selection and dimensionality reduction techniques.

- Filter methods: Statistical techniques, such as chi-square tests, analysis of variance (ANOVA)[50], and F-ratios [79], are used to identify relevant features. Methods such as correlation-based feature selection (CFS)[79] and maximum-relevance minimum-redundancy (mRMR) [80] maximize relevance and minimize redundancy.
- Wrapper/embedded methods: Algorithms such as recursive feature elimination (RFE)[81], [82] and genetic algorithms (GA)[83] refine features iteratively. Regularization techniques, such as least

absolute shrinkage [73], the selection operator (Lasso) [73], and ElasticNet [84] optimize feature sets.

- Dimensionality reduction: Techniques such as principal component analysis (PCA) [85], linear discriminant analysis (LDA) [46], and T-distributed stochastic neighbor embedding (t-SNE)[86] simplify data while preserving variance or class separability.
- Optimization algorithms: Advanced techniques such as Particle Swarm Optimization (PSO) [75], [87], the Adaptive Controlling Parameter-based African Vultures Optimization Algorithm (ACP-AVOA) [88], and the Multi-Agent Salp Swarm Algorithm (MASS) [47] optimize high-dimensional feature spaces to improve model performance.

2) ALGORITHMS FOR CLASSIFICATION AND SCREENING

Table 5 in Supplementary Material E categorizes speech and language disorders, the datasets used, and the corresponding algorithms, highlighting advancements in leveraging AI for identifying and classifying diverse impairments. Support vector machines (SVMs) [89], random forests (RFs) [90], [91] and k nearest neighbors (KNNs)[92] are effective for smaller datasets and less complex tasks. For example, SVMs [89] perform well with datasets such as NEMOURS for dysarthria, whereas RF and KNN [93] are commonly used for articulation [94] and pronunciation disorders. Deep learning architecture, including deep neural networks (DNNs)[95], convolutional neural networks (CNNs)[96], [97], and recurrent neural networks (RNNs), dominate speech and language disorder classification. For example, DNNs and CNNs effectively extract high-dimensional features for dysarthria classification via datasets such as TORGO and UA-Speech. Advanced techniques such as bidirectional long short-term memory (BiLSTM) networks and transformer-based models have been applied to neurodegenerative conditions such as Parkinson's disease and Alzheimer's disease.

Hybrid models and ensemble methods increase robustness and generalizability. Combining stacked ensembles with ML[10] and DL[98] algorithms is used for Parkinson's and vocal cord disorders, which perform better for classification. Hybrid ANN-HMM and hybrid deep ensemble models have been employed for stuttering and fluency disorders. These methods exploit the advantages of different algorithms by integrating them into the same system. This technique improves accuracy by combining the temporal features captured by the RNNs and the spatial features extracted by CNNs. Other methods, such as the mixture-of-experts (MoE)[99], transfer learning and goodness-of-pronunciation (GoP)[100] methods with neural networks, have demonstrated potential in conditions such as Parkinson's disease and dysarthria. The trend toward deep learning approaches reflects their adaptability in handling high-dimensional data, which is beneficial for complex disorders. Emerging multimodal techniques leverage diverse inputs, such as spectrographic data, linguistic annotations, and

articulatory signals, to better capture the multidimensional aspects of speech impairments. Spatial convolutional neural networks (SCNNs) with multihead attention transformers (MHATs) [101] and graph-based learning are promising for multimodal datasets.

Results and trends from Table 4 of Supplementary Material D: Analysis of methodologies across disorders and datasets reveals key trends:

- Dysarthria: Neural network models such as CNNs, DNNs, and LSTMs are widely used, leveraging the temporal nature of symptoms with datasets such as TORGO and UA-Speech.
- Parkinson's Disease: Methods range from traditional ML models such as SVM to transformer-based sequence-to-sequence models, supported by datasets such as PC-GITA and EEG.
- Stuttering: CNN-BiLSTM combinations and hybrid models effectively capture spatial and temporal variations via datasets such as FluencyBank and SEP-28k.
- Alzheimer's Disease: SVM and DNN models dominate, particularly with datasets such as NCMMSC2021, efficiently identifying acoustic features linked to cognitive decline.
- Pronunciation and Articulation Disorders: Articulation datasets such as TCC-300 rely on simpler methods such as Apriori, whereas pronunciation error detection benefits from GRU-RNN architectures.
- Voice Disorders: CNNs, SL-RANs, and light CNNs analyze datasets such as MEEI and Saarbruecken, reflecting the complexity of these conditions.
- Fluency and Speech Intelligibility: Fluency disorders benefit from time-delay neural networks (TDNNs) and hybrid classifiers, whereas intelligibility classification leverages DNNs and SVMs with datasets such as UA-Speech.
- Emerging conditions: Although the primary focus of this review is on AI techniques across general speech and language disorders, we recognize the growing relevance of pediatric conditions such as Specific Language Impairment (SLI). Several recent studies have explored age-aware and hybrid deep learning models—like SLINet and CNN-LSTM architectures—which achieved high diagnostic accuracies on child speech datasets such as LAANA. These efforts highlight the applicability of AI methods in addressing developmental variability and early screening challenges in pediatric populations.

EVALUATION METRICS

To evaluate model performance in speech processing and classification tasks, a wide range of metrics are employed, each offering insights based on specific goals, data characteristics, and application contexts. These metrics include accuracy measures, error rates, and correlation-based metrics. Accuracy-based metrics evaluate the proportion of correctly predicted instances and are widely used for binary and multiclass classification tasks. Accuracy measures the ratio of true positives and true negatives to the total number of predictions, making it a standard metric, although metrics such as unweighted accuracy are needed for imbalanced datasets [69]. Classification accuracy is commonly used in

tasks such as speech recognition, providing an overall measure of model performance [102]. The recognition rate further refines this evaluation by focusing on the percentage of accurately identified speech units, highlighting a model's precision in specific recognition tasks [52].

Error rates are critical for assessing transcription accuracy and intelligibility. The word error rate (WER) calculates insertion, deletion, and substitution errors at the word level, making it a key metric for transcription tasks [51]. Similar metrics include the phoneme error rate (PER) and the character error rate (CER), which focus on phonemes and characters, respectively, and are particularly useful for languages with complex structures [103], [104]. The syllable error rate (SER) measures errors per syllable and can be used to evaluate intelligibility and clarity, especially in rehabilitation applications [105]. The precision, recall, and F1 score are critical evaluation metrics for classification models. The precision is the number of true positives compared with all the predicted positive cases. This is significant in applications where the occurrence of false positives must be minimized, such as in diagnostics [64]. Recall, known as sensitivity, is defined as the fraction of true positives out of all actual positive cases to avoid missing cases [64]. The F1 score, which is the harmonic mean of precision and recall, provides a balanced measure that is especially useful in imbalanced datasets [49], [85].

Specificity and balanced metrics provide improved evaluations of model performance. Specificity measures the proportion of correctly identified negatives, complementing sensitivity to assess discriminatory power [73]. Balanced accuracy (BAC) averages sensitivity and specificity, mitigating the effects of class imbalance and providing a more equitable measure of performance [76]. Similarly, G-Mean balances sensitivity and specificity through a geometric mean, making it valuable for evaluating imbalanced datasets [106]. The area under the curve (AUC) metrics assess a model's ability to distinguish between classes across thresholds. AUC-ROC evaluates this distinction by plotting true positive rates against false positive rates, with higher values indicating robust performance [107]. AUC-PRC focuses on precision and recall, providing additional insights into scenarios where these measures are prioritized [83]. Correlation-based metrics assess the relationship between the predicted and actual values. In continuous or ordinal predictions, Pearson and Spearman correlation coefficients are widely used, as they show linear and monotonic relationships [77]. For binary classification problems, the Matthews correlation coefficient (MCC) is used. It considers all true and false predictions and is a well-known measure in an imbalanced dataset [108].

Mean error metrics are applied in regression tasks to quantify the average magnitude of errors. The mean absolute error computes the average of the absolute differences between the predictions and the actual values, whereas the root mean square error (RMSE) penalizes large deviations, making it a

better choice for applications in which large errors are undesirable [80], [109]. The average mean absolute error (MAE) is an improvement that considers different contexts of prediction [80]. Other metrics and ratios provide specialized insights for speech processing. Unweighted average recall (UAR) calculates recall across all classes without weights for class frequency, making it valuable for multiclass and imbalanced datasets [63]. The penalized expected deviance (PED) and detection cost function (DCF) are commonly applied in security and authentication tasks to penalize errors on the basis of application risk [110], [111]. Confusion matrix derivatives, such as the true positive rate (TPR), false positive rate (FPR), and equal error rate (EER), are used to evaluate biometric and security applications where a balance between false positives and negatives is critical.

D. RQ4: APPLICATIONS AND ASSISTIVE TECHNOLOGIES

RQ4 explores current applications of AI-based speech and language disorder classification in various domains with their limitations. This review identifies different applications, such as teleradiology and telemonitoring systems, early detection and noninvasive screening [47], assistive technology for speech impairments [50], multimodal and advanced diagnostic tools [49], telehealth and real-time monitoring [51], automated pathology detection and classification [39], language learning and pronunciation training [88], [112], therapeutic tools and speech rehabilitation [113], smart healthcare and edge computing applications [114], clinical decision support systems [72], mobile health [115] and low-cost diagnostic solutions [90]. For Parkinson's disease, teleradiology [71] and telemonitoring [99] reduce outpatient visits while supporting real-time monitoring and therapy adjustments. Dysarthria-focused tools improve communication through assistive systems and early diagnosis [61], [62]. Voice disorders benefit from diagnostic tools in mobile and real-time settings [102], [115]. Tools for stuttering, SLI (Speech Language Impairment), and aphasia provide targeted assessments, aiding therapy [65].

E. RQ5: SPEECH ENHANCEMENT METHODS

Table 6 in supplementary material F summarizes recent speech enhancement studies focused on improving intelligibility, quality, and downstream processing of speech in noisy or impaired conditions. These studies deploy diverse AI models across multiple datasets and real-world scenarios. One of the prominent contributions is the MFFR-Net model [15], which combines multi-scale feature fusion with attention recalibration. Evaluated on the TIMIT and LibriSpeech datasets, this model yielded a 17.3% improvement in STOI, illustrating the benefits of integrating spectral and temporal features for intelligibility enhancement.

The SRU-based lightweight recurrent model [116] leverages simplified recurrent units to support real-time deployment,

achieving an improvement of +0.69 in Perceptual Evaluation of Speech Quality (PESQ) and +0.19 in Short-Time Objective Intelligibility (STOI). This model demonstrates that low-latency solutions can maintain high enhancement accuracy, particularly for embedded or mobile speech applications. In parallel, the Spiking-FullSubNet [16] introduces neuromorphic computing into speech enhancement by integrating self-organizing maps and spiking neural networks (SOM-SNN). Evaluated using Scale-invariant signal-to-distortion ratio (SI-SDR) and Deep Noise Suppression Mean Opinion Score (DNSMOS), it targets ultra-low-power use cases such as hearing aids and assistive communication devices.

Other models adopt self-supervised learning or spatial awareness techniques. For example, the Noise-Aware Self-Supervised Variational Autoencoder (VAE) [14] enhances intelligibility using NISQA and STOI-NET metrics, while the SINA-MONaural model [117] exploits spatial information to boost quality in monaural speech enhancement contexts. The Nested U-Net with channel attention [118] employs convolutional architectures to optimize performance under unseen noise conditions, showing robustness in practical deployment scenarios.

Importantly, one study applied GAN-based architectures [119] to convert whispered to voiced speech for laryngeal patients, extending enhancement beyond noise removal to functional speech restoration. Similarly, a hybrid dysarthric speech enhancement model [120] focused on correcting articulation distortions, directly targeting communication impairments in speech disorders.

Across these studies, common evaluation metrics included PESQ, STOI, SI-SDR, and DNSMOS. However, a critical observation is the limited validation of these enhancement methods on disordered or pathological speech corpora—most being trained on generic or synthetically noisy datasets. Despite this, the performance improvements and versatility across applications (e.g., hearing aids, voice assistants, speech therapy tools) affirm the relevance of enhancement techniques in AI-based assistive systems.

VI. DISCUSSION

In RQ1, speech and language disorders encompass a wide spectrum of conditions, each with distinct challenges for AI-driven classification and diagnosis. Among the 16 disorders reviewed, Parkinson's disease and dysarthria emerged as the most studied, accounting for 59 and 32 papers, respectively. This focus likely reflects the availability of well-structured datasets, the prevalence of these conditions, and the clear clinical need for advanced diagnostic and monitoring tools. Similarly, voice disorders, addressed in 53 papers, demonstrate the field's emphasis on improving voice pathology assessments and noninvasive diagnostic solutions. Neurodegenerative disorders differ fundamentally from other speech and language disorders because of their progressive nature, leading to significant temporal variability in speech patterns. This deterioration significantly impacts the design and application of AI models, datasets, and

practical implementations for speech and language processing. This multifaceted nature complicates speech modeling. Disease progression varies significantly between individuals and is influenced by factors such as age, comorbidities, and treatment interventions. The progressive nature of neurodegenerative disorders requires AI systems that go beyond static classification to support continuous monitoring and adaptation. The variability in individual disease trajectories, influenced by age, comorbidities, and treatment responses, can be assisted by personalization.

In response to RQ2, the predominant obstacle encountered in the classification process is the insufficiency of accessible speech corpora, as the majority are proprietary, thereby impeding researchers from effectively training and evaluating their frameworks. Among the 212 scholarly articles selected, 28 pertained to English corpora, and 19 addressed corpora in other languages. Table 5 shows that neurodegenerative research is limited by a lack of diverse, longitudinal datasets, which are critical for tracking disease progression, whereas studies of other disorders face issues such as data imbalance and overlapping symptoms. This comparison highlights RQ3, the need for adaptive and explainable AI for neurodegenerative disorders and broader dataset development for other disorders.

Visual cues such as facial expressions, lip movements, and gestures are crucial, especially in conditions such as dysarthria, Parkinson's disease, or aphasia, where motor and articulatory symptoms are evident in both audio and visual modalities. Therefore, integrating video features with speech [121] could improve performance in classification tasks and enhance interpretability by revealing how visible speech cues are related to detected disorders.

While answering RQ3, Table 7 in supplementary material shows that for neurodegenerative disorders such as Parkinson's disease, Alzheimer's disease, and dysarthria, advanced methods such as CNNs, LSTMs, and DNNs are more frequently employed. These techniques effectively address the temporal and multimodal complexities of progressive speech impairments. In contrast, some disorders, such as stuttering, articulation disorders, and fluency disorders, have a broader distribution of approaches, and some of them rely on simpler methods, such as KNN, SVM, and decision trees. These methods align with the static nature of these conditions and the availability of focused datasets.

Feature extraction/selection algorithms enhance model accuracy by reducing overfitting, training time and dimensional space. Section 4 lists the different feature extraction and selection algorithms for extracting acoustic, linguistic, and auxiliary features. The reviewed studies demonstrate that both machine learning and deep learning algorithms are widely used for classification tasks. Machine learning techniques are widely applied when datasets are limited. These models are generally shallow (limited layers

or complexity) and rely on human-engineered features such as acoustic measurements or linguistic scores. They often offer faster training and more transparent decision-making than deep learning models do, although they may not capture complex patterns as effectively. Among machine learning algorithms, the support vector machine (SVM) [93] model stands out for its robust performance. SVMs achieve high accuracy, on the order of ~85–95% in binary classification tasks, given quality features and balanced data. A linear SVM[122] can offer moderate interpretability by highlighting which features contribute most to the decision boundary. However, nonlinear SVMs act as black boxes, making their decisions less transparent to clinicians. Overall, traditional ML models excel when feature engineering is strong and data are limited, offering faster deployment and some level of transparency at the expense of the highest possible accuracy.

In the realm of deep learning, several architectures have shown high accuracy in classifying different speech and language disorders, as they automatically learn feature representations from raw data. Convolutional neural networks (CNNs) [123] are used for analyzing acoustic signals by converting them to time–frequency representations such as spectrograms. CNNs learn local patterns such as formant structures or turbulence noise that are indicative of disorders. CNN-based architecture has shown high accuracy, often reaching 90% accuracy in distinguishing disordered vs. healthy speech in controlled settings; however, CNNs remain black boxes for clinicians. RNNs are designed to capture temporal dependencies in sequential data. In speech disorder tasks, RNN models [124] are used to analyze time series of acoustic features or sequences of words/phonemes. An RNN can learn the progression of a speaker's articulation through a sentence or the pattern of sound repetitions in stuttering. These models are deep when stacked in multiple layers, can also be combined with CNNs, and yield improved accuracy by capturing both spectral and temporal cues. However, RNNs suffer from low interpretability. The learned memory states

and gating values do not map cleanly onto human-understandable features.

Transformer-based architectures use self-attention mechanisms to capture long-range dependencies in data. In speech disorder classification, transformers are being explored in two main ways: (1) Audio transformers, such as spectrogram-based transformers, and (2) Language transformers applied to speech transcripts or linguistic features. The model can be combined with a CNN, which performs well in classification. As the models are very deep, they require large datasets or pretraining. The attention mechanism in transformers does offer a crumb of interpretability; one can inspect attention weight distributions to determine which parts of the input the model focuses on.

As shown in Table 2, the shallow models such as SVM and RF depend on carefully chosen acoustic/linguistic features and tend to be easier to interpret, whereas deep models such as CNNs, RNNs, and transformers automatically learn complex patterns from data and often reach higher accuracy in terms of interpretability. There is a clear trade-off between model complexity and transparency. The choice of model also correlates with the disorder type and data modality. A static classification such as “PD vs healthy” from a sustained vowel might be effectively handled by an SVM on a small feature set, whereas a nuanced task such as “assess dysarthria severity from a continuous speech passage” may benefit from a deep CNN-LSTM that captures subtle spectro-temporal cues.

Notably, models incorporating the short-time Fourier transform (STFT) with layered convolutional neural networks (CNNs), standalone CNNs, hybrid models combining artificial neural networks (ANNs), multilayer long short-term memory (LSTM) networks, and bidirectional long short-term memory (BiLSTM) models have proven effective. Transfer learning approaches that use LSTM and GRU models, along with customized CNN architectures, achieve high accuracy across various speech and language disorder classification tasks.

TABLE 2
STRUCTURED COMPARISON OF REPRESENTATIVE MODELS

Model	Disorders	Feature Types	Model Complexity	Performance	Clinical Interpretability
Support Vector Machine (SVM)[85]	Parkinson's Disease, Dysarthria, Voice Pathologies, Stuttering	Acoustic and some linguistic	Shallow	High for binary tasks (80–95% accuracy)	Moderate
Random Forest (RF)[46]	Parkinson's, Aphasia, Voice Disorders	Acoustic; Linguistic; Prosodic (hand-	Shallow Ensemble	High on structured data (~80–90%). Handles	Moderate

		crafted feature sets)		overfitting by averaging.	
GMM-HMM (Gaussian Mixture/Hidden Markov Model)[120]	Dysarthria, Voice Disorders, Fluency (Stuttering)	Acoustic sequential features (spectra, cepstra over time)	Complex due to per-speaker acoustic tuning, tied-state optimization.	Moderate	Low
Convolutional Neural Network (CNN)[117]	Dysarthria, Voice Disorders, Parkinson's	Acoustic (spectrogram images, raw audio waveforms); augmented linguistic features as additional input	Deep – multiple convolutional layers + pooling	90% accuracy	Low
Recurrent Neural Network (RNN)/LSTM [118]	Stuttering, Dysarthria, Parkinson's (prosody)	Acoustic time-series (frame-by-frame features); Linguistic sequences (transcripts)	Deep – recurrent layers (LSTM/GRU units) often stacked, sometimes with CNN front-end	High	Low
Transformer-Based Model[96]	Voice Disorders, Dysarthria, Cognitive-Communication disorders	Acoustic, Linguistic, Multimodal fusion of both	Very Deep – multihead attention layers, large parameter counts(pretrained models)	Robust performance	Low

Table 3 illustrates the key insights from our findings that can be used in future research and clinical applications in AI-driven speech disorder classification. This table captures novel perspectives that emerged across model families. Foundational metrics such as accuracy, F1 score, and AUC-ROC were widely reported across the reviewed studies, reflecting their utility in general classification tasks. For specific challenges, such as imbalanced datasets in neurodegenerative disorders, metrics such as the Matthews correlation coefficient (MCC) [125] allow balanced evaluation across classes. Additionally, tasks involving

transcription accuracy, such as fluency and intelligibility assessments, use metrics such as the word error rate (WER).

As stated in RQ4, AI-driven speech assistive technology can enhance communication capabilities for individuals with speech and language disorders through adaptive automatic speech recognition systems. AI-based classification approaches for speech and language disorders have been deployed across a range of application domains, including telediagnosis[126], telemonitoring [71], early detection, and therapeutic intervention. These systems facilitate noninvasive, real-time assessment,

Table 3
Key Insights and Implications from Comparative Analysis of Classification Models

Insight/Interpretation	Implication	Model(s)
Hybrid models balance accuracy and interpretability	Combining deep feature extractors (CNNs) with interpretable classifiers (RF, decision trees)	CNN+RF, CNN+rule-based systems
Multimodal inputs enhance generalization	Models using acoustic, linguistic, and physiological features outperform unimodal models	CNN-RNN-Transformer hybrids, Multimodal DNNs
Underuse of transformer models in clinical settings	Transformer models (AST, BERT) are rarely used clinically due to explainability issues and data demands	Transformer-based architectures
Temporal modeling potential is underexplored	LSTM and RNN could be beneficial beyond fluency disorders (e.g., dysarthria, Parkinson's disease)	RNNs, LSTMs
Clinical interpretability decreases with model depth	Shallow models preferred clinically; deeper models powerful but less interpretable	All deep models (CNNs, RNNs, Transformers)

Generalizability linked to linguistic diversity of datasets	Cross-linguistic training critical for robustness in broader clinical applications	All models, notably SVM and CNN
Voice datasets larger than language datasets	Larger voice datasets (e.g., MEEI) allow DL adoption; small datasets limit DL use for language disorders (e.g., aphasia, SLI)	Transformers & DL models (aphasia, SLI)
Self-supervised learning is underutilized	Self-supervised models (Wav2Vec) underexplored in clinical speech domains	Self-supervised Transformers
Limited adoption of explainability tools	Explainability (SHAP, attention maps) underrepresented but crucial for clinical trust	All DL models
Shift toward continuous monitoring	Models increasingly used for continuous disorder monitoring; rising importance of mobile/edge applications	Lightweight CNNs, LSTMs (mobile devices)

improve accessibility to care, and alleviate the demands on clinical infrastructure, particularly in underserved or resource-limited settings. Assistive technologies have demonstrated substantial utility in enhancing communicative abilities among individuals with disorders such as dysarthria, stuttering, and aphasia while also contributing to rehabilitation and the delivery of personalized therapeutic strategies. Nevertheless, key challenges persist in the areas of model generalizability, effective multimodal integration, and rigorous clinical validation.

While multimodal approaches offer promise, their effectiveness is constrained by the need for standardized and annotated multimodal datasets. Future efforts must address these limitations to enable robust modeling and generalizability across disorders. The review identified a notable gap in the use of spontaneous speech data for diagnosing speech disorders. Most of the reviewed papers highlighted the need to develop and share standardized, high-quality pathological voice databases to encourage reproducibility and collaborative progress across research teams. Moreover, several studies highlight the need to generalize models to real-life settings by validating their performance on unseen data and addressing language-dependent acoustic variations. The incorporation of personalized assistive systems for various conditions can also enhance practical usability. The review discussed the need to incorporate a broader range of acoustic features, such as prosody, pitch, and timbre, alongside advanced spectro-temporal features, to improve detection accuracy. The integration of wearable sensors and cloud-based platforms was also emphasized as a promising direction for continuous voice monitoring, especially in telemedicine and remote assessments. Addressing linguistic and cultural diversity remains critical, with a gap in models that are cross linguistically validated. Additionally, leveraging large language models (LLMs) could enable disorder-agnostic frameworks, providing a unified approach to manage diverse speech pathologies while addressing crosslinguistic variability.

In addition to performance accuracy, computational efficiency and real-time feasibility are essential for integration into assistive technologies, especially for mobile

health applications and edge devices. While some models, such as STFT-layered CNNs, achieve high accuracy (up to 99.89%) and BiLSTM (94.1%), their computational complexity and memory requirements may restrict deployment on low-power devices. In contrast, approaches utilizing GMM-HMM, SVM, or hand-crafted features such as MFCCs offer faster inference with lower energy consumption, although often with slightly reduced accuracy. Few studies reported metrics like inference latency, model size, or energy usage, indicating a gap in evaluating deployability trade-offs. Deep spectro-temporal models and Transformers offer excellent recognition performance but typically require cloud support or GPU acceleration. Optimizing models through compression, quantization, or lightweight architectures remains critical for achieving real-time, on-device applications.

This review identified that AI models have shown significant promise in advancing classification, diagnosis, and therapeutic support for individuals with speech and language disorders. Across the reviewed papers, it is evident that while classification remains a primary focus, speech enhancement is emerging as a critical complementary area, particularly in the context of real-world and clinical deployment.

RQ5 underscores the growing role of AI-driven speech enhancement techniques in improving the intelligibility and perceptual quality of disordered or noisy speech. Enhancement methods such as multi-scale feature fusion networks (e.g., MFFR-Net), spiking neural networks (Spiking-FullSubNet), lightweight recurrent units (SRU), and GAN-based whispered-to-voiced conversion models not only contribute to improved user experience but also enable more accurate downstream classification in systems designed for disorder detection and monitoring. However, one key observation is that many enhancement models are trained predominantly on clean or artificially corrupted corpora, with limited validation on pathological or clinically diverse datasets. This discrepancy highlights the need for enhancement models to be adapted or retrained for real-world pathological speech conditions to ensure robust clinical application.

From a clinical translation perspective, integrating enhancement models as a preprocessing step in speech

disorder diagnostic pipelines offers tangible benefits, such as reduced background noise, improved clarity for ASR engines, and enhanced patient-clinician interaction, particularly in telemedicine and assistive contexts. Despite promising advances, speech enhancement remains underrepresented in current literature when compared to classification. As AI-enabled speech interfaces increasingly enter clinical and home-based settings, enhancement will be critical for generalizability, usability, and accessibility, especially for individuals with severe impairments or speech affected by environmental factors.

Furthermore, embedding enhancement architectures in real-time, low-latency systems, such as mobile health platforms or wearable devices, aligns well with trends toward continuous monitoring and edge computing. Lightweight models such as Simple Recurrent Unit (SRU) or transformer-based speech enhancement architectures, when optimized for energy efficiency and response time, can support a broader ecosystem of AI-enabled assistive technologies. In the future, combining enhancement and classification in a unified multimodal framework will provide an end-to-end, robust solution for diagnosing and managing speech and language disorders.

In summary, the discussion of RQ5 highlights that while classification remains the cornerstone of current AI research, speech enhancement plays a critical role in improving diagnostic performance, communication quality, and user engagement—thus serving as both a complementary and enabling technology in the broader AI-driven speech disorder landscape.

A. IMPLICATIONS FOR PRACTICE AND POLICY

The following implications can be used to help build AI systems for neurodegenerative disorders so that they are more adaptive, inclusive, and ethical, addressing the needs of patients, clinicians, and researchers:

Dynamic AI models for evolving needs: AI systems designed for neurodegenerative disorders must be capable of adapting to the progressive nature of these conditions. Techniques such as lifelong learning and adaptive modeling help models remain effective as patients' symptoms evolve, while incorporating explainability ensures that clinicians can trust and understand AI decisions, particularly in monitoring disease progression.

Limited datasets: A significant obstacle in tackling illnesses such as MS and LBD is the scarcity of publicly accessible and longitudinal datasets. In contrast to PD, which has advantages over well-organized datasets such as PC-GITA and the UCI repository, MS and LBD encounter difficulties in dataset development due to episodic symptoms and overlapping speech impairments, emphasizing the necessity for teamwork to create multimodal datasets that accurately reflect complicated illness development.

Moreover, AI models for illnesses such as myasthenia gravis must consider symptom variability affected by treatment responses, introducing an additional layer of complexity.

Longitudinal datasets: Documenting temporal alterations in speech resulting from neurodegeneration necessitates varied and longitudinal datasets. These datasets must include consistent recordings of natural speech and task-specific prompts from the same participants for longitudinal progress analysis. Moreover, to enhance generalizability, datasets should encompass varied linguistic and cultural contexts to facilitate cross-cultural and multilingual applications.

Personalization for patient-specific care: The advancement of neurodegenerative diseases differs markedly among individuals, requiring customized AI systems. Multimodal methodologies that integrate acoustic and linguistic characteristics are inadequately employed in certain illnesses, such as specific language impairment (SLI) and stuttering, which could benefit from tailored instruments for improved results. These systems must incorporate carer and physician feedback to customize interventions, whereas mobile and telehealth platforms are employed for real-time progress monitoring and adaptive treatment modification.

Managing ethical challenges with public datasets: Publicly available datasets are vital for advancing AI research in neurodegenerative disorders but raise significant privacy and security concerns. Ensuring privacy and minimizing bias in speech data remain significant challenges owing to the scarcity of publicly accessible, well-annotated datasets. Anonymization techniques such as differential privacy or synthetic data, tiered access models, and dynamic consent systems can safeguard patient identities and empower them to control their data, ensuring ethical and effective research. **Explainable and ethical AI in practice:** Models used for integrating AI into clinical workflows need to be both explainable and ethical. Transparent systems build trust with clinicians and facilitate sensible decisions. Enforcing ethics committees can ensure that datasets and models are developed based on high standards of privacy and inclusivity.

B. FUTURE WORK

To address the challenges and gaps identified in this review, the following areas represent future directions for advancing AI in speech and language disorder classification.

Creating comprehensive datasets: Future research needs longitudinal, multimodal datasets capturing the temporal progression of speech impairment across diverse demographics and languages. The lack of multimodal datasets represents a significant research gap and an opportunity for interdisciplinary collaboration. Future work should prioritize collecting and curating such datasets while ensuring ethical data handling, anonymization, and accessibility. Cross-cultural datasets enhance AI generalizability, ensuring inclusive diagnostics, whereas variations in speaking styles and noise conditions improve adaptability to real-world scenarios.

Development of adaptive AI systems: speech and language disorders, particularly neurodegenerative disorders, show

variability and progression; therefore, adaptive AI models are needed. Techniques such as lifelong learning and transfer learning can be pursued to allow AI systems to adapt dynamically to individual patient trajectories, enhancing clinical relevance and personalization. Moreover, reinforcement learning can be used to train interactive AI models that refine their predictions based on clinician feedback, thus creating a dynamic loop between the AI system and medical experts.

Integration of multimodal data: Future research should use wearable sensors that encompass acoustic, linguistic, and physiological data alongside real-time speech processing. The multimodal framework facilitates a comprehensive understanding of speech and language impairments, enhancing diagnosis and therapeutic strategies. Advanced multimodal fusion techniques, such as attention-based transformers or joint embedding spaces, should be explored to enhance heterogeneous data type integration.

Explainable and ethical AI: Explainability is critical for integrating AI into clinical workflows. Future studies should focus on developing transparent models that clinicians can interpret and trust. Several explainable AI techniques have been developed to address these concerns, including:

SHAP (SHapley Additive exPlanations) which provides feature-level explanations based on cooperative game theory, indicating the contribution of each input feature to a model's output. It is particularly useful for understanding acoustic or articulatory features that influence classification decisions.

LIME (Local Interpretable Model-Agnostic Explanations) perturbs the input space locally to approximate the behavior of black-box models, making it suitable for model-agnostic interpretability in speech-based classification tasks.

Grad-CAM (Gradient-weighted Class Activation Mapping) helps visualize the spatial importance of spectrogram features in classification tasks related to voice disorders.

In addition, ethical considerations such as privacy, informed consent, and equitable dataset usage should be the guiding principles in advancing AI to ensure responsible deployment.

Expanding research on underrepresented conditions: high incidence but underrepresented disorders such as multiple sclerosis, Lewy body dementia, and myasthenia gravis, and therefore, research must be focused. The development of condition-specific datasets and AI models will help to fill the gap in the current literature and provide tools for diagnosing and managing those conditions. Using unsupervised pretraining methods under data-scarce conditions improves model performance with a decrease in dependence on large, annotated datasets.

Validation across languages and cultures: Validation across languages and cultures: AI systems need to be rigorously cross linguistically and culturally validated to ensure their effectiveness in diverse populations. This approach will enhance the global applicability of speech and language diagnostic tools, addressing disparities in healthcare access. Transfer learning and federated learning can be particularly

effective for adapting models to low-resource languages without requiring extensive local datasets.

Advancing clinical applications: the integration of AI systems into clinical practice should focus on telemonitoring, assistive technologies, and rehabilitation tools. This will ensure that practical and scalable solutions for managing speech and language disorders are developed through collaborative partnerships between researchers, clinicians, and industry stakeholders. Real-time speech monitoring tools using edge computing can reduce latency and make AI systems function effectively in resource-constrained environments.

AI techniques: Transformer architectures and large pretrained models incorporate knowledge learned from massive datasets, which can be fine-tuned to detect subtle anomalies in disordered speech. Specifically, large language models (LLMs) such as ChatGPT and Whisper are increasingly being applied to extract higher-level semantic and syntactic patterns from speech. For instance, LLMs can be leveraged to generate dynamic speech transcripts for clinical documentation or interact with patients as real-time dialogue agents for preliminary screening. These systems can also support symptom progression monitoring by correlating temporal linguistic features with disease trajectories.

Development of pathology-specific enhancement models: Current enhancement techniques are predominantly trained on general-purpose or synthetic noise datasets. Future research should prioritize training and validating enhancement models on pathological speech corpora to ensure performance generalization across diverse disorder types such as dysarthria, stuttering, or aphasia. These models must be sensitive not only to background noise but also to the unique spectral distortions introduced by vocal pathologies.

Fusion of enhancement with classification frameworks: Instead of treating enhancement and classification as independent modules, future systems should adopt end-to-end architectures that jointly optimize enhancement and classification. This co-training can ensure that the enhancement process preserves clinically relevant features while improving clarity and intelligibility, particularly for ASR-based diagnostic systems.

Low-resource and real-time enhancement systems: There is a critical need for lightweight and energy-efficient enhancement models suitable for deployment on mobile or embedded devices. Architectures such as Spiking Neural Networks, efficient transformers, and quantized RNNs show potential for use in low latency telemonitoring and wearable platforms, enabling real-time feedback in remote care scenarios.

Multimodal enhancement strategies: Future work should explore multimodal enhancement, where visual cues (e.g., lip movement, facial expressions) complement degraded audio signals to improve restoration accuracy. This is particularly valuable for disorders with articulatory

disruptions, such as apraxia or severe dysarthria. Multimodal architectures like SpeechT5 and CLIP illustrate the ability to align audio, visual, and textual inputs, which is particularly useful for disorders involving articulatory impairments or facial motor dysfunctions.

Innovative approaches such as graph neural networks and federated learning provide effective solutions for handling intricate, distributed information while maintaining privacy and security. Moreover, self-supervised learning and generative models, including diffusion models, can improve performance by utilizing unlabeled speech data and generating synthetic data to address the insufficiency of databases for uncommon illnesses.

VII. CONCLUSION

The synthesized research on AI applications for speech and language disorder classification addressed key research questions, highlighting gaps and advancements in the field. For RQ1, a disproportionate focus on specific conditions, such as Parkinson's disease, is driven largely by the availability of datasets tailored to these disorders. In contrast, high incidence yet underrepresented conditions, including multiple sclerosis and Lewy body dementia, remain insufficiently studied, highlighting a misalignment between research focus and clinical need. Addressing this imbalance is essential for advancing equitable AI applications in speech pathology.

For RQ2, dataset limitations emerged as a critical barrier. Current datasets often lack longitudinal, multimodal, and crosslinguistic diversity, constraining the generalizability of AI models. The review emphasized that future advancements in dataset creation must focus on capturing the temporal progression of speech impairments across diverse populations and environments. Inclusive datasets from diverse languages, such as speech and language disorders, will help generalize applications to various populations.

For RQ3, the identification and classification of speech and language disorders constitute a highly dynamic domain of research. However, there is notable reliance on CNN and DNN models, along with traditional features such as MFCCs. Few studies have used multimodal and advanced deep learning approaches. Neurodegenerative disorders pose unique challenges due to their progressive nature. Future research requires dynamic AI models such as self-supervised learning capable of tracking temporal changes in speech patterns and adapting to individual progression.

For RQ4, the applications of AI in speech and language disorders are promising, particularly in diagnostics, telemonitoring, and rehabilitation. However, challenges in clinical adoption remain, including the limited interpretability of models and the need for real-world validation. Addressing these issues requires the integration of explainable AI techniques, such as SHAP values and counterfactual explanations, alongside ethical frameworks to ensure privacy, transparency, and equitable deployment. [2]

For RQ5, speech enhancement has emerged as a crucial yet underrepresented area in the domain of AI-based speech disorder systems. Enhancement techniques, particularly those employing deep learning architecture such as CNNs, BiLSTMs, and transformer-based models, have shown potential in improving speech clarity and intelligibility in pathological speech. These methods are especially valuable when used as preprocessing modules to improve classification accuracy or are integrated into real-time assistive technologies. However, few studies address the pathology-specific nuances required for effective enhancement, signaling the need for disorder-adapted enhancement models and better evaluation metrics tailored to impaired speech. Future systems should prioritize the co-optimization of enhancement and classification tasks in a unified pipeline to better reflect real-world deployment scenarios.

AI has transformative potential in diagnosing, monitoring, and managing neurodegenerative speech and language disorders. Leveraging advanced AI techniques and multidisciplinary collaboration can yield highly accurate, personalized, accessible diagnostic and assistive technologies, thereby substantially improving quality of life. Addressing the key challenges of data availability, model interpretability, and privacy concerns is crucial for fully realizing the benefits of AI in clinical practice. The findings highlight the need for a shift in AI research for speech and language disorders, prioritizing the creation of comprehensive datasets, the development of adaptive and multimodal models, and the application of novel AI techniques to underrepresented conditions and the integration of speech enhancement as a foundational step in building reliable speech technology for impaired populations.

Credit authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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